Chia-Hsiang Kao

Ph.D. in Computer Science, Cornell University · Expected Fall 2027

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Google Scholar · GitHub

RESEARCH INTERESTS

I build world models that infer and predict the hidden dynamics of real-world interactions: recovering 3D motion and physics from observation, and generating controllable video.

- **3D & physical world modeling:** completing occluded scene geometry and motion, and recovering rigid-body physics from video [CVPR 2026].
- **Video generation & in-context learning:** unifying video tasks through spatiotemporal analogy [under review].
- **LLM agents:** code-executing, tool-using agents and their evaluation [ACL 2026 Findings].
- **AI for science:** remote sensing [NeurIPS 2024 D&B], DNA modeling [ICML 2024, ICLR 2024 Workshop], and medical imaging [MICCAI 2021, EMBC 2022].

EDUCATION

Ph.D. in Computer Science | Cornell University, Ithaca, NY — 2023 – Fall 2027 (expected)
Advisor: Bharath Hariharan

Doctor of Medicine (M.D.) | National Yang-Ming Chiao-Tung University, Taipei, Taiwan — 2015 – 2022

RESEARCH EXPERIENCE

Student Researcher | Meta, New York, NY — Jan 2026 – May 2027
Host: Menglin Jia

- **Amodal scene flow completion (ongoing):** leading scene-level completion of occluded geometry together with its 3D motion, beyond prior visible-only scene flow; designed a novel point-cloud tokenizer and a diffusion model that completes point-voxel tokens while predicting the motion field.
- **ViGeo (Preprint):** led a video in-context learning framework that unifies video tasks as spatiotemporal canvas completion; developed a single model that absorbs a diverse task taxonomy spanning camera control, modality-guided generation, and video manipulation, and generalizes to unseen tasks and sensing modalities; fully fine-tuned Wan2.2-5B on 32 A100 nodes.

Applied Scientist Intern | Amazon, Sunnyvale, CA — May 2025 – Aug 2025
Host: Chien-Yi Wang

- **Δynamics (CVPR 2026):** led a VLM that recovers rigid-body physics from monocular video by generating MuJoCo scene configurations as structured text. Trained on 400K synthetic videos with natural-language motion reasoning as auxiliary supervision, it substantially outperforms frontier VLMs, improves further with test-time search, and transfers to a new benchmark of 235 real-world videos.

Graduate Researcher | Cornell University, Ithaca, NY — 2023 – Present
Advisor: Bharath Hariharan

- **UnivEarth (ACL 2026 Findings):** led a benchmark of 140 Earth-observation questions across 17 satellite sensors for code-executing LLM agents using the Google Earth Engine API; showed frontier agents reach only 33% accuracy, with generated code failing 58% of the time.
- **AllClear (NeurIPS 2024 D&B):** led a dataset and benchmark for cloud removal in satellite imagery.
- **Counter-current learning (NeurIPS 2024):** proposed a biologically plausible alternative to backpropagation using anti-parallel feedforward and feedback networks.

MENTORING

- Hangyu Zhou: AllClear (NeurIPS 2024 Datasets & Benchmarks, co-first author)
- Cheryl Lam and Aarush Umap: UnivEarth (ACL 2026 Findings)

SKILLS

Modeling: world models, video generation (flow matching, Wan2.2), VLMs, LLM agents & tool use
Tools: PyTorch, FastVideo, multi-node distributed training (32 A100 nodes), MuJoCo, Google Earth Engine

PUBLICATIONS

Google Scholar: 500+ citations, h-index 8 (Oct 2026)

2026 **Unifying Video Tasks via Spatiotemporal Analogy**
Under review
Chia-Hsiang Kao, Belinda Zeng, Bharath Hariharan, Menglin Jia

Δynamics: Language-Based Representation for Inferring Rigid-Body Dynamics From Videos
CVPR 2026
Chia-Hsiang Kao, Cong Phuoc Huynh, Chien-Yi Wang, Noranart Vesdapunt, Stefan Stojanov, Bharath Hariharan, Oleksandr Obiednikov, Ning Zhou

2025 **Towards LLM Agents for Earth Observation**
ICML 2025 TerraBytes Workshop; ACL 2026 Findings
Chia-Hsiang Kao, Wenting Zhao, Cheryl Lam, Aarush Umap, Shreelekha Revankar, Samuel Speas, Snehal Bhagat, Rajeev Datta, Cheng Perng Phoo, Utkarsh Mall, Carl Vondrick, Kavita Bala, Bharath Hariharan

2024 **Counter-Current Learning: A Biologically Plausible Dual Network Approach for Deep Learning**
NeurIPS 2024
Chia-Hsiang Kao, Bharath Hariharan

AllClear: A Comprehensive Dataset and Benchmark for Cloud Removal in Satellite Imagery
NeurIPS Datasets and Benchmarks Track 2024
Hangyu Zhou*, Chia-Hsiang Kao*, Cheng Perng Phoo, Utkarsh Mall, Bharath Hariharan, Kavita Bala

Caduceus: Bi-Directional Equivariant Long-Range DNA Sequence Modeling
ICML 2024
Yair Schiff, Chia-Hsiang Kao, Aaron Gokaslan, Tri Dao, Albert Gu, Volodymyr Kuleshov

2023 **Advancing DNA Language Models: The Genomics Long-Range Benchmark**
AAAI 2023 Workshop; ICLR 2024 MLGenX Workshop
Chia-Hsiang Kao*, Evan Trop*, McKinley Polen*, Yair Schiff*, Bernardo P. de Almeida, Aaron Gokaslan, Thomas Pierrot, Volodymyr Kuleshov

FedBug: A Bottom-Up Gradual Unfreezing Framework for Federated Learning
Preprint
Chia-Hsiang Kao, Yu-Chiang Frank Wang

2022 **MEG-Based Classification and Grad-CAM Visualization for Major Depressive and Bipolar Disorders with Semi-CNN**
IEEE EMBC 2022
Chun-Chih Huang, Intan Low, Chia-Hsiang Kao, Chuan-Yu Yu, Tung-Ping Su, Jen-Chuen Hsieh, Yong-Sheng Chen, Li-Fen Chen

MAML Is a Noisy Contrastive Learner in the Classification
ICLR 2022
Chia-Hsiang Kao, Wei-Chen Chiu, Pin-Yu Chen

2021 **Demystifying T1-MRI to FDG18-PET Image Translation via Representational Similarity**
MICCAI 2021
Chia-Hsiang Kao, Yong-Sheng Chen, Li-Fen Chen, Wei-Chen Chiu

HONORS & AWARDS

- Foundational AI PhD Fellowship, Cornell University — 2025
- Student Travel Award, MICCAI — 2021
- Undergraduate Research Fellowship, National Science and Technology Council, Taiwan — 2018, 2020
- Summer Research Fellowship, National Health Research Institutes, Taiwan — 2018

SERVICE

Conference Reviewer: NeurIPS (2021, 2024–2026), ICLR (2025–2027), ICML (2025), ECCV (2026), COLM (2026), AAAI (2025), AISTATS (2025), AutoML (2022)

Journal Reviewer: CVIU (2022), Computers & Electrical Engineering (2024), IEEE TETCI (2024)